

EXPERIMENT NO 18

Determination and plotting of Antenna Half power beamwidth.

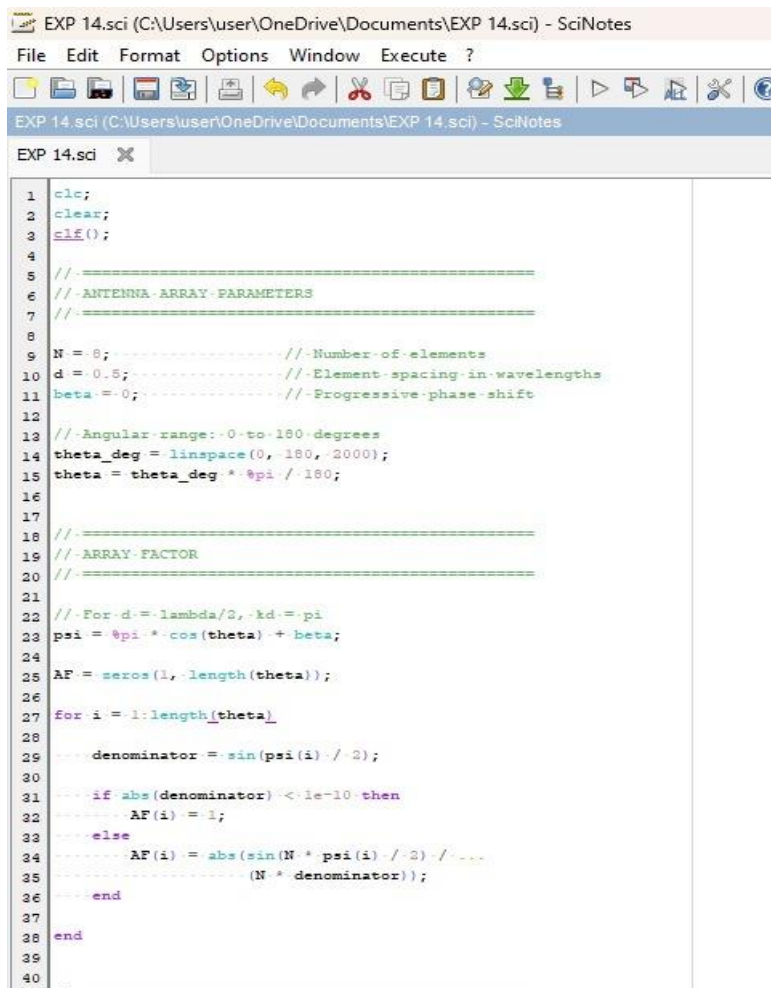
Aim:

To determine and plot Half Power Beamwidth (HPBW) and to determine the angular width at -3 dB points

Software required:

- SCILAB version 6.0.1

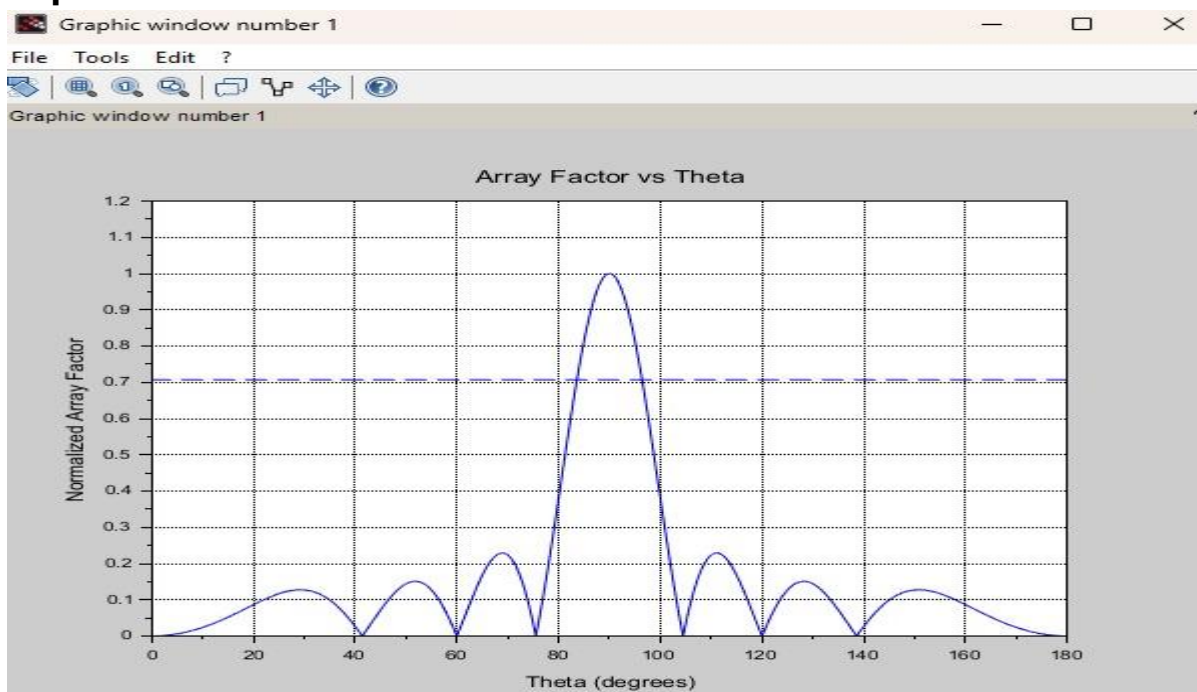
Program:



```
1  clc;
2  clear;
3  clf();
4
5  // =====
6  // -ANTENNA-ARRAY-PARAMETERS
7  // =====
8
9  N = 8; // -Number-of-elements
10 d = 0.5; // -Element-spacing-in-wavelengths
11 beta = 0; // -Progressive-phase-shift
12
13 // -Angular-range:-0-to-180-degrees
14 theta_deg = linspace(0, 180, 2000);
15 theta = theta_deg * %pi / 180;
16
17 // =====
18 // -ARRAY-FACTOR
19 // =====
20
21 // -For d = lambda/2, kd = pi
22 psi = %pi * cos(theta) + beta;
23
24 AF = zeros(1, length(theta));
25
26 for i = 1:length(theta)
27
28     denominator = sin(psi(i) / 2);
29
30     if abs(denominator) < 1e-10 then
31         AF(i) = 1;
32     else
33         AF(i) = abs(sin(N * psi(i) / 2) /
34                     (N * denominator));
35     end
36 end
37
38 end
39
40
```

```
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
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EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
EXP 14.sci X
69
70 end
71
72
73 //=====
74 // DISPLAY HPEW
75 //=====
76
77 mprintf("\n===== \n");
78 mprintf("----- HALF-POWER-BEAMWIDTH- (HPEW) \n");
79 mprintf("===== \n");
80 mprintf("Number of Elements = %d \n", N);
81 mprintf("Element Spacing = %.2f * lambda \n", d);
82 mprintf("HPEW = ..... = %.3f -degrees \n", HPEW);
83 mprintf("===== \n");
84
85
86 //=====
87 // PLOT ARRAY FACTOR
88 //=====
89
90 figure(1);
91
92 plot(theta_deg, AF);
93
94 xlabel("Theta (degrees)");
95 ylabel("Normalised Array Factor");
96 title("Array Factor vs Theta");
97
98 xgrid();
99
100 a = gca();
101 a.data_bounds = [0, 0; 180, 1.05];
102
103
104 //=====
105 // SHOW HALF-POWER LEVEL
106 //=====
107
108 plot(theta_deg, threshold * ones(theta_deg), "--");
109
```

Output:



RESULT: The experiment was done and the Half power beamwidth of the given antenna was calculated. Different case studies were done.